Senthen Velmurugan

Dallas, TX · Open to relocation anywhere in the US (Pacific Northwest / Texas preferred)
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Education

Georgia Institute of Technology **In Progress**
Master of Science in Computer Science (OMSCS) *Atlanta, GA*

The University of Texas at Dallas **Completed**
Bachelor of Science in Computer Science *Dallas, TX*

Experience

KloudGin **May 2024 – August 2024**
Machine Learning Engineering Intern *Remote*

- ◇ Raised first-contact resolution 20% over 12 weeks of use by taking the Generative AI team's open question, whether RAG could speed up customer case resolution, from proposal to production rollout
- ◇ Gave support engineers suggested next actions from similar past cases and accepted forum solutions by building a multi-source RAG system: an ingestion pipeline, hybrid vector + BM25 retrieval, and cross-encoder re-ranking
- ◇ Raised top-k precision by measuring retrieval offline and through an A/B feedback loop with support agents, then diagnosing an overfit to long threads and retuning chunking and metadata filters
- ◇ Let other teams integrate it safely with tenant-scoped search, PII scrubbing, safe prompting, and a documented API

GBC Computers **May 2023 – August 2023**
Cybersecurity and Systems Intern *Mabank, TX*

- ◇ Improved client network performance by monitoring with PRTG and tracing bottlenecks in Wireshark captures
- ◇ Hardened client infrastructure with pfSense firewall rules, Snort intrusion detection, and endpoint protection
- ◇ Found and patched critical risks with Nessus vulnerability assessments, validated by OpenVAS and manual audits

Projects

P&ID Digitizer: Scanned Engineering Drawings to Graphs | *Python, PyTorch, OpenCV, FastAPI, Docker* **2026**

- ◇ Turned scanned P&IDs into queryable graphs (GraphML, DEXPI 2.0.1 XML) at 0.915 edge F1 and 0.997 symbol mAP@0.5 on a 75-sheet test split by chaining symbol detection, line extraction, and graph assembly
- ◇ Found small symbols on 7,000 px sheets by fine-tuning a Faster R-CNN on overlapping 1024^2 tiles, merged across seams
- ◇ Sent only doubtful edges to a person by calibrating per-edge confidence for a review UI, shipped as a FastAPI job service in a non-root Docker image with strict mypy and 99 pytest cases in CI

DEXPI P&ID Viewer | *JavaScript, Python, XML, SVG* **2026**

- ◇ Let anyone open a DEXPI 2.0.1 plant model with nothing installed or uploaded by building a zero-dependency, single-page viewer that lays out diagrams from topology alone, with no stored coordinates
- ◇ Caught invalid files before downstream tools do by validating structure and the metamodel in the browser and a CLI

Word2Eq: Seq2Seq Training & Serving | *Python, PyTorch, FastAPI, Docker, Prometheus* **2024 – In Progress**

- ◇ Took valid equations from 97.4% to 100% on 1,000 SVAMP problems by rebuilding a non-functional prototype into a train, evaluate, and serve platform for a 6.4M-parameter Transformer with grammar-constrained decoding
- ◇ Served 150 requests/s at 73 ms p95 on one laptop CPU by micro-batching behind a bounded queue that sheds overload with a fast 503, and made training preemption-safe with bit-identical resumes

Campbell Woodworks: Storefront for a Family Woodshop | *Jekyll, Liquid, HTML, CSS, Python* **2026**

- ◇ Put a family woodshop's work, once only in Facebook posts behind a login, in front of buyers at no running cost
- ◇ Covered 23 pieces with one Markdown file each, and call buttons instead of a cart, since the shop sells by phone

Thai Sizzle: Website for a Family Food Truck | *Jekyll, Liquid, HTML, CSS, Python* **2026**

- ◇ Put a Thai food truck's menu, prices, hours, and location online, with all 13 dishes matched to its menu board
- ◇ Made it owner-editable with one data file per section, plus a weekly GitHub Action that refreshes the Google rating

Additional Projects

Metal-to-Pod: Self-Hosted CI/CD to Kubernetes | *Kubernetes, Bash, Go, Docker, Jenkins* **2026 – In Progress**

- ◇ Took every commit to a running pod with no cloud services on 4 VMware VMs (16 GB laptop) by building a kubeadm cluster with default-deny network policy, a git hook into Jenkins as code, and a private TLS registry
- ◇ Made releases safe to automate by gating on unit tests, shipping ∼10 MB distroless non-root images tagged by commit, and rolling back automatically when an external smoke test fails
- ◇ Ran a Go delivery-promise service against 99.5% availability and 300 ms latency SLOs, with burn-rate alerts unit-tested in promtool, 8 failure drills, and 5 runbooks

Personal VPN | *Terraform, Go, Bash, WireGuard, Linux* **2026 – In Progress**

- ◇ Kept traffic on untrusted networks private without trusting a commercial VPN by provisioning a hardened WireGuard server on Hetzner for about €4/month with one `terraform apply`
- ◇ Kept every private key off the network and out of Terraform state by generating keys on each device and the server key on first boot, managed with `vpnctl`, a Go CLI with 11 unit tests
- ◇ Cut the attack surface to one UDP port with default-deny nftables and cloud firewall rules, and documented a threat model of what it does and does not protect

Honeypot & Attack Map | *Terraform, AWS, Bash, Docker, Cowrie* **2026 – In Progress**

- ◇ Collected attacker sessions without letting the bait become a launch point by running a Cowrie SSH/Telnet honeypot in an isolated AWS VPC from Terraform, with capabilities dropped and all egress denied except S3
- ◇ Kept every event safe and queryable by shipping it off-box with Vector to a versioned, write-only S3 bucket as partitioned NDJSON for Athena or DuckDB, with budget alerts on spend

The $2 Billion Patch Note: Market Event Study | *Python, pandas, NumPy, SciPy, Matplotlib* **2026 – In Progress**

- ◇ Measured how Counter-Strike 2's October 2025 trade-up update repriced about 3,900 items (a 19% drawdown, trough at 25 hours) by building price indices from daily Steam market medians
- ◇ Showed newly useful Covert inputs gained 43.5% mean abnormal return within 48 hours ($t = 8.95$) with a constant-mean event study by rarity and wear, in a tested Python CLI that regenerates every figure

The Limits of Foresight: Depth-Limited Reasoning in Go and Poker | *Python, LaTeX* **2026 – In Progress**

- ◇ Designed two small, interpretable models with one shared depth parameter to test whether people stop thinking ahead the same way about board positions (Go) and other minds (poker)
- ◇ Set four testable hypotheses to fit to public human game data, for a pitch milestone in CS 6795 (Cognitive Science) at Georgia Tech, fall 2026; results still to come

Personal Website: sent-hen.github.io | *Jekyll, Liquid, HTML, CSS, JavaScript* **2023 – In Progress**

- ◇ Published a personal site with nothing to run or maintain as a static Jekyll site on GitHub Pages, where every list is one data file, so adding a project is a one-file change
- ◇ Drew its animations (0/1 rain, golden spiral, Sierpinski triangle, ripple) as generated SVG and plain JavaScript with no libraries, with a Hi/Lo switch for still line art

The Thousand and One Words: SAT Vocabulary Comic Epic | *Python, HTML, SVG* **2026 – In Progress**

- ◇ Made 1,000 SAT words learnable as one story by writing a 100-chapter comic epic for readers aged 14 to 16, ten words a chapter, each with a story scene, a picture link, and a memory-palace station
- ◇ Generated all 100 chapter pages, image prompts, and the combined script from one Python source, so a one-line edit updates every page

Technical Skills

Languages: Python, JavaScript, Go, C/C++, SQL, Bash, HTML, CSS, LaTeX
Frameworks: PyTorch, FastAPI, LangChain, OpenCV, pandas, NumPy, SciPy, NetworkX, pytest, Jekyll
Tools: Linux (Debian), Git, Docker, Kubernetes, Terraform, PostgreSQL, Pinecone, CI/CD, Autodesk Fusion, Jenkins, Prometheus, AWS, GitHub Actions, WireGuard, nftables, pfSense, Snort, Nessus, Wireshark

Organizations

Mensa **2013 – Present**